READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

How a prehistoric predator took to the skies



Is that a bird in the sky? A plane? No, it's a pterosaur. Kate Thomas meets Professor Matthew Wilkinson, who built a life-size model to find out how this prehistoric predator ever got off the ground.

Pterosaurs existed from the Triassic period, 220 million years ago, to the end of the Cretaceous period, 65 million years ago, when South America pulled away from Africa and the South Atlantic was formed. They are among the least understood of all the extinct reptiles that once spent their lives in the skies while the dinosaurs dominated the land. Pterosaurs had no feathers, but at least part of their bodies was covered in hair, not unlike bats. Some believe this is an indication they were warm-blooded. Researchers also debate whether pterosaurs travelled on the ground by walking on their hind legs, like birds, or by using all fours, relying on their three-toed front feet as well as their four-toed rear feet.

Pterosaurs were vertebrates, meaning they were the first species possessing backbones to become airborne, but scientists have never quite understood their flight technique. How, they wondered, did such a heavy creature ever manage to take off? How could a wing that appears to have been supported by fine, hollow bones have carried one into the sky? Then came the discovery of a site in Brazil's Araripe basin. Here, not only were hundreds of fossils of amphibians* and other reptiles found, but archaeologists unearthed a number of very well-preserved pterosaurs. The anhanguera — a fish-eating subspecies of pterosaur that ruled the skies in the Cretaceous period — was among them. With a wingspan of up to 12 metres, they would have made an amazing sight in the sky — had any human been there to witness it. 'I've been studying pterosaurs for about eight years now,' says Dr Matthew Wilkinson, a professor of zoology at Cambridge University. With an anhanguera fossil as his model, Wilkinson began gradually reconstructing its skeletal structure in his Cambridge studio. The probability of finding three-dimensional pterosaur fossils anywhere is slim. 'That was quite a find,' he says. 'Their bones are usually crushed to dust.' Once the structure was complete, it inspired him to make a robot version as a way to understand the animal's locomotion. With a team of model-makers, he has built a remote-controlled pterosaur in his studio. 'Fossils show just how large these creatures were. I've always been interested in how they managed to launch themselves, so I thought the real test would be to actually build one and fly it.'

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* amphibians: animals that can live both in water and on land.

Wilkinson hasn't been alone in his desire to recreate a prehistoric beast. Swiss scientists recently announced they had built an amphibious robot that could walk on land and swim in water using the sort of backbone movements that must have been employed by the first creatures to crawl from the sea. But Wilkinson had the added complication of working out his beast's flight technique. Unlike those of bats or flying squirrels, pterosaur wings — soft, stretchy membranes of skin tissue — are thought to have reached from the chest to the ankle, reinforced by fibres that stiffened the wing and prevented tearing. Smaller subspecies flapped their wings during takeoff. That may have explained the creatures' flexibility, but it did not answer the most pressing question: how did such heavy animals manage to launch themselves into the sky? Working with researchers in London and Berlin, Wilkinson began to piece together the puzzle.

It emerged that the anhanguera had an elongated limb called the pteroid. It had previously been thought the pteroid pointed towards the shoulder of the creature and supported a soft forewing in front of the arm. But if that were the case, the forewing would have been too small and ineffectual for flight. However, to the surprise of many scientists, fossils from the Araripe basin showed the pteroid possibly faced the opposite way, creating a much greater forewing that would have caught the air, working in the same way as the flaps on the wings of an aeroplane. So, with both feet on the ground, the anhanguera might have simply faced into the wind, spread its wings and risen up into the sky. Initial trials in wind tunnels proved the point — models of pterosaurs with forward-facing pteroids were not only adept at gliding, but were agile flyers in spite of their size. 'This high-lift capability would have significantly reduced the minimum flight speed, allowing even the largest forms to take off without difficulty,' Wilkinson says. 'It would have enabled them to glide very slowly and may have been instrumental in the evolution of large size by the pterosaurs.'

Resting in the grass at the test site near Cambridge, the robot model's wings ripple in the wind. In flight, the flexible membrane, while much stiffer than the real thing, allows for a smooth takeoff and landing. But the model has been troubled by other mechanical problems. 'Unlike an aircraft, which is stabilised by the tail wing at the back, the model is stabilised by its head, which means it can start spinning around. That's the most problematic bit as far as we're concerned,' Wilkinson says. 'We've had to take it flying without the head so far.' When it flies with its head attached, Wilkinson will finally have proved his point.

So what's next for the zoologist — perhaps a full-size Tyrannosaurus rex? 'No,' he tells me. 'We're desperate to build really big pterosaurs — creatures with even greater wingspans, weighing a quarter of a ton. But,' he adds, just as one begins to fear for the safety and stress levels of pilots landing nearby at Cambridge City Airport, 'it's more likely we'll start off with one of the smaller, flapping pterosaurs.' This is certainly more reassuring. Let's hope he is content to leave it at that.

Questions 27–32

Complete the summary using the list of words, ***A–L****, below.*

Write the correct letter, ***A–L****, in boxes 27–32 on your answer sheet.*

Pterosaurs are believed to have existed until the end of the Cretaceous period. They are classed as **27** ……, which were capable of flight, although, unlike modern species, they had some **28** ……, which is evidence of their having had warm blood. There are two theories as to how they moved on land: perhaps with all their feet or by using their **29** …… only. Another mystery has concerned the ability of the pterosaur to fly despite its immense **30** ……, and the fact that the bones making up the wing did not have great **31** …… . Thanks to reptile fossils found in Brazil, we now know that the subspecies known as anhanguera had wings that were 12 metres across and that it mainly survived on **32** …… .

A	front feet	**B**	fish	**C**	dinosaurs
D	reptiles	**E**	flexibility	**F**	hind legs
G	amphibians	**H**	birds	**I**	strength
J	weight	**K**	tail	**L**	hair

Questions 33–36

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes 33–36 on your answer sheet, write

YES	*if the statement agrees with the claims of the writer*
NO	*if the statement contradicts the claims of the writer*
NOT GIVEN	*if it is impossible to say what the writer thinks about this*

33 It is rare to find a fossil of a pterosaur that clearly shows its skeleton.

34 The reason for building the model was to prove pterosaurs flew for long distances.

35 It is possible that pterosaur species achieved their wing size as a result of the pteroid.

36 Wilkinson has made several unsuccessful replicas of the pterosaur's head.

Questions 37–40

Choose the correct letter, ***A, B, C*** *or* ***D****.*

Write the correct letter in boxes 37–40 on your answer sheet.

37 What was Professor Wilkinson's main problem, according to the third paragraph?

A Early amphibians had a more complex structure than pterosaurs.
B Pterosaur wings could easily be damaged while on the ground.
C Flying squirrels and bats were better adapted to flying than pterosaurs.
D Large pterosaurs were not able to take off like other flying animals.

38 What did Professor Wilkinson discover about a bone in pterosaurs called a pteroid?

A It was in an unexpected position.
B It existed only in large species of pterosaurs.
C It allowed pterosaurs to glide rather than fly.
D It increased the speed pterosaurs could reach in the air.

39 According to the writer, the main problem with the remote-controlled 'pterosaur' is that

A it has been unable to leave the ground so far.
B it cannot be controlled when its head is attached.
C its wing material is not flexible enough.
D the force of the wind may affect its test results.

40 What does 'it' in the last sentence refer to?

A the information the tests have revealed
B Wilkinson's sense of achievement
C Wilkinson's desire to build models
D the comparison between types of models

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Questions 27–32 选词填空详细解析

题号	答案	题干翻译	定位原文与翻译	详细解释
27	**D reptiles**	翼龙被认为一直存在到白垩纪末期。它们被归类为 **27……**，这些动物能够飞行。	**第 1 段**：They are among the least understood of all the **extinct reptiles** that once spent their lives in the skies…译文：它们属于所有曾经生活在天空中的**已灭绝爬行动物**中最不为人了解的一类。	空格前是 **are classed as**，意思是“被归类为”。原文直接说翼龙是 **extinct reptiles**，所以选 **reptiles 爬行动物**。注意不能选 dinosaurs，因为原文说 dinosaurs dominated the land，而翼龙是在天空生活的 extinct reptiles，并不是恐龙。
28	**L hair**	虽然不像现代物种那样，它们有一些 **28……**，这是它们拥有温血的证据。	**第 1 段**：Pterosaurs had no feathers, but at least part of their bodies was covered in **hair**, not unlike bats. Some believe this is an indication they were warm-blooded.译文：翼龙没有羽毛，但它们身体至少有一部分覆盖着**毛发**，与蝙蝠有些相似。有些人认为这表明它们是温血动物。	题干中的 **evidence of their having had warm blood** 对应原文 **an indication they were warm-blooded**。原文说明“温血”的证据是身体覆盖着 **hair**，所以答案是 **L hair**。
29	**F hind legs**	关于它们如何在陆地上移动有两种理论：也许是用所有的脚，或者只用它们的 **29……**。	**第 1 段**：Researchers also debate whether pterosaurs travelled on the ground by walking on their **hind legs**, like birds, or by using all fours…译文：研究人员还争论翼龙在地面上移动时，是像鸟一样用**后腿**行走，还是四足行走……	题干中的 **with all their feet** 对应原文 **using all fours**；题干中的 **by using their … only** 对应原文 **walking on their hind legs**。所以填 **hind legs 后腿**。
30	**J weight**	另一个谜团是：尽管翼龙有巨大的 **30……**，它们却仍然能够飞行。	**第 2 段**：How, they wondered, did such a **heavy** creature ever manage to take off?译文：他们想知道，这样一种**沉重的**生物究竟是如何起飞的?	题干中的 **despite its immense…** 对应原文 **such a heavy creature**。heavy 表示“重的”，名词形式对应选项 **weight 重量**。所以答案是 **J weight**。
31	**I strength**	另一个谜团还包括：组成翅膀的骨头并没有很强的 **31……**。	**第 2 段**：How could a wing that appears to have been supported by fine, hollow bones have carried one into the sky?译文：一只似乎由细而中空的骨头支撑的翅膀，怎么能够把这样的动物带上天空呢?	原文说翼龙翅膀由 **fine, hollow bones** 支撑，即“细而中空的骨头”，这暗示骨头看起来不够结实。题干改写为 **the bones … did not have great…**，应填 **strength 强度 / 力量**。所以选 **I strength**。
32	**B fish**	多亏了在巴西发现的爬行动物化石，我们现在知道，被称为 anhanguera 的亚种翼展达 12 米，并且主要以 **32……** 为生。	**第 2 段**：The anhanguera — a **fish-eating** subspecies of pterosaur… With a wingspan of up to 12 metres…译文：anhanguera 是翼龙的一个**食鱼**亚种……翼展可达 12 米。	题干中的 **mainly survived on** 意思是“主要以……为食 / 生存”。原文说 **fish-eating**，即“吃鱼的”，所以答案是 **B fish**。

Questions 33–36 判断题详细解析

题号	答案	题干翻译	定位原文与翻译	详细解释
33	**YES**	很少能发现清楚显示翼龙骨骼结构的化石。	**第 2 段**：The probability of finding three-dimensional pterosaur fossils anywhere is slim. ‘That was quite a find,’ he says. ‘Their bones are usually crushed to dust.’ 译文：在任何地方找到三维翼龙化石的可能性都很小。“那真是一个了不起的发现，” 他说，“它们的骨头通常会被压碎成粉末。”	题干说 **It is rare to find…**，原文说 **probability … is slim**，意思是 “可能性很小”。题干中的 **clearly shows its skeleton** 对应原文中的 **three-dimensional pterosaur fossils** 和 **bones are usually crushed to dust**。原文支持题干，所以答案是 **YES**。
34	**NO**	建造模型的原因是为了证明翼龙能够长距离飞行。	**第 2 段**：Once the structure was complete, it inspired him to make a robot version as a way to understand the animal’s locomotion… I’ve always been interested in how they managed to launch themselves…译文：结构完成后，这启发他制作一个机器人版本，以此来了解这种动物的运动方式……我一直对它们如何起飞感兴趣……	题干说建模型是为了证明翼龙 **flew for long distances 长距离飞行**。但原文说目的是了解 **locomotion 运动方式**，尤其是 **how they managed to launch themselves 它们如何起飞**。原文目的不是证明长距离飞行，因此与原文矛盾，答案是 **NO**。
35	**YES**	翼龙物种可能是由于 pteroid 这种骨头而发展出了它们的翅膀尺寸。	**第 4 段**：…the pteroid possibly faced the opposite way, creating a much greater forewing…以及：It would have enabled them to glide very slowly and may have been instrumental in the evolution of large size by the pterosaurs.译文：pteroid 可能朝向相反的方向，从而形成了大得多的前翼……这会使它们能够非常缓慢地滑翔，并且可能对翼龙大型化的进化起到了重要作用。	题干中的 **It is possible** 对应原文 **may have been**，都表示 “可能”。原文说 pteroid 的朝向形成了更大的前翼，并且这种高升力能力可能推动了翼龙大型化的进化。题干说翼龙的翼展 / 翅膀尺寸可能与 pteroid 有关，与原文一致，所以答案是 **YES**。
36	**NOT GIVEN**	Wilkinson 已经制作了几个不成功的翼龙头部复制品。	**第 5 段**：…the model is stabilised by its head, which means it can start spinning around. That’s the most problematic bit… We’ve had to take it flying without the head so far.译文：这个模型靠头部来保持稳定，这意味着它可能开始旋转。这是目前最让我们头疼的问题……到目前为止，我们不得不让它在没有头部的情况下飞行。	原文只说 “带着头飞会有稳定性问题，所以目前只能不带头飞”，但没有说 Wilkinson 做了 **several 几个** 头部复制品，也没有说这些复制品都 **unsuccessful 不成功**。题干信息无法从原文判断，所以答案是 **NOT GIVEN**。

Questions 37–40 选择题详细解析

题号	答案	题干翻译	定位原文与翻译	详细解释
37	**D**	根据第三段，Wilkinson 教授的主要问题是什么？	**第 3 段**：But Wilkinson had the added complication of working out his beast's flight technique... Smaller subspecies flapped their wings during takeoff... but it did not answer the most pressing question: how did such heavy animals manage to launch themselves into the sky?译文：但是 Wilkinson 还面临一个额外的复杂问题，即弄清楚这种动物的飞行技术……较小的亚种在起飞时会拍打翅膀……但这并没有回答最紧迫的问题：如此沉重的动物是如何把自己送上天空的?	第三段的核心问题是：**大型、沉重的翼龙如何起飞**。选项 D "Large pterosaurs were not able to take off like other flying animals." 意思是 "大型翼龙不能像其他飞行动物那样起飞"，概括了原文中 "较小亚种可以拍翅起飞，但大型沉重翼龙如何起飞仍是问题"。所以选 **D**。A 提到早期两栖动物结构更复杂，原文只是举瑞士科学家机器人例子，不是 Wilkinson 的主要问题。B 说翅膀在地面容易损坏，原文未说这是主要问题。C 说飞鼠和蝙蝠更适合飞行，原文只是比较翅膀结构，不是主要问题。
38	**A**	Wilkinson 教授发现翼龙中一种叫 pteroid 的骨头有什么特点？	**第 4 段**：It had previously been thought the pteroid pointed towards the shoulder... However, to the surprise of many scientists, fossils from the Araripe basin showed the pteroid possibly faced the opposite way...译文：人们之前认为 pteroid 指向翼龙的肩部……然而，令许多科学家惊讶的是，Araripe 盆地的化石显示 pteroid 可能朝向相反的方向。	原文中的 **to the surprise of many scientists** 表明这个发现出乎意料；**faced the opposite way** 说明它的位置 / 朝向与以前认为的不同。A "It was in an unexpected position." 意为 "它处在一个出乎意料的位置"，正确。B "只存在于大型翼龙" 未提及。C "让翼龙只能滑翔而不是飞行" 错误，原文说它们不仅善于滑翔，还很灵活。D "提高飞行速度" 错误，原文说它降低了最低飞行速度。
39	**B**	根据作者，遥控 "翼龙" 的主要问题是什么？	**第 5 段**：Unlike an aircraft, which is stabilised by the tail wing at the back, the model is stabilised by its head, which means it can start spinning around. That's the most problematic bit... We've had to take it flying without the head so far.译文：飞机是由后面的尾翼保持稳定的，而这个模型是由头部保持稳定的，这意味着它可能开始旋转。那是我们目前最头疼的问题……到目前为止，我们不得不让它在没有头部的情况下飞行。	原文说模型带头飞时可能 **spinning around 旋转**，所以他们目前只能 **without the head 不带头** 试飞。B "it cannot be controlled when its head is attached" 意为 "当头部装上时，它无法被控制"，符合原文。A 错：原文说它能 smooth takeoff and landing。C 错：膜虽然比真实翼龙硬，但仍允许顺利起飞和降落。D 错：原文没有说风力会影响测试结果。
40	**C**	最后一句中的 "it" 指什么？	**最后一段**：'We're desperate to build really big pterosaurs... But... it's more likely we'll start off with one of the smaller, flapping pterosaurs.' This is certainly more reassuring. Let's hope he is content to leave it at that.译文："我们非常想建造真正巨大的翼龙模型……不过……我们更可能会先从一种较小的、会拍翅的翼龙开始。" 这当然更令人放心。希望他满足于做到这一步。	**leave it at that** 是固定表达，意思是 "到此为止 / 不再进一步"。这里作者希望 Wilkinson 不要继续发展到更大、更吓人的模型，而是满足于建造较小的翼龙模型。因此 **it** 指 Wilkinson 继续建造模型的愿望或计划。选项 C "Wilkinson's desire to build models" 最符合。A "测试揭示的信息" 不是最后一句所指。B "成就感" 原文未提。D "不同模型类型之间的比较" 不是 it 的指代对象。